5	(a)	A simple a.c. generator which consists of a rectangular coil with 40 turns rotates at 50 revolutions per second in a uniform magnetic field of flux density 50 mT. The coil is 50 cm long and 35 cm wide, as shown in Figure 5.1. An electromotive force (e.m.f.) is induced in the coil.

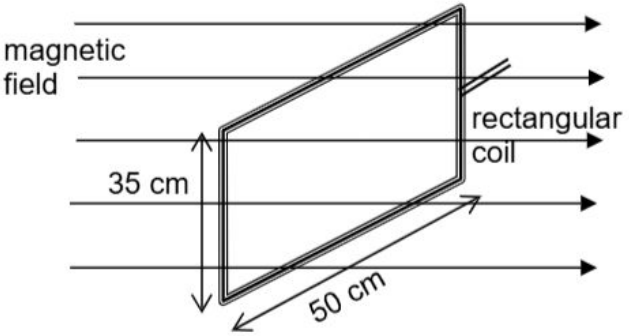


Figure 5.1

The maximum induced e.m.f. E_o in the coil is given by the expression

E_o = maximum magnetic flux linkage × angular speed of the coil

Use this expression to calculate the maximum e.m.f. induced E_o.

E_o = ...V [2]

L1

$E_o = \Phi_{max} \times \omega$

$= NBA \times 2\pi f$

= (40) (50 x 10^{-3}) (50 x 10^{-2})(35 x 10^{-2}) × (2π)(50) **M1**

= 109.96 = **110 V (Accept 2 s.f. or 3 s.f.)** **A1**

(b) The output from the generator in part **(a)** is connected to the input of a transformer. The transformer has 40 turns on its primary coil and 10 turns on its secondary coil. The transformer may be considered to be ideal. The secondary coil is connected in series to a resistor of resistance 100 Ω and a diode, as shown in Figure 5.2.

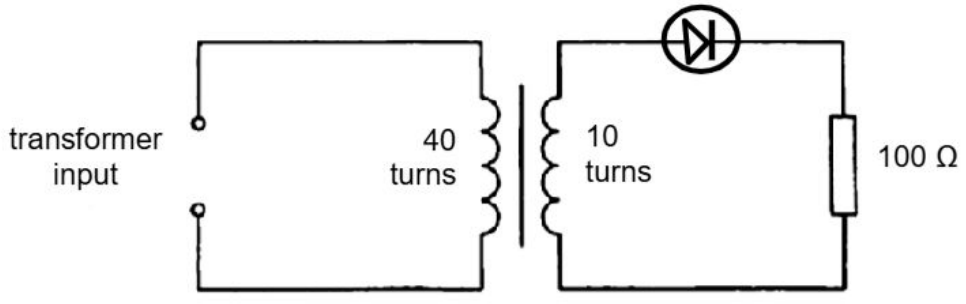


Figure 5.2

Calculate the root-mean-square voltage across the 100 Ω resistor.

root-mean-square voltage = ...V [3]

	L3	Let V_0 be the **peak** voltage across the resistor. $$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$ $$\frac{\boldsymbol{V_0}}{\mathbf{109.96}} = \frac{\mathbf{10}}{\mathbf{40}}$$ V_0 = 27.49 V Since the voltage across the resistor is **half-wave rectified**, $\langle V^2 \rangle = \frac{V_o^2}{4}$ $\therefore \sqrt{\langle V^2 \rangle} = \frac{V_o}{2} = \frac{\mathbf{27.49}}{\mathbf{2}}$ = 13.745 = **14 V (Accept 2 s.f. or 3 s.f.)**	**M1** **M1** **A1**

[Total: 5]